Classical Mechanics

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Chapter 1. Newton mechanics

$$\sum_i \mathbf{F}_i = m\ddot{\mathbf{x}}. \tag{1.1}$$

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Chapter 2. Lagrangian mechanics

[E1] The possible physical states in Lagrangian mechanics

We need first to identify the configuration space of the system, which is an S-dimensional manifold $\mathcal{Q}$, encoding the position information of the system. For example, for a system of N particles moving in three-dimensional space without constraints, the configuration space is $\mathbb{R}^{3N}$. For the rigid body, the configuration space is $\mathbb{R}^3 \times SO(3)$. But specially, for a rigid thin rod, there exists an $SO(2)$ symmetry, so the configuration space is $\mathbb{R}^3 \times \frac{SO(3)}{SO(2)} \cong \mathbb{R}^3 \times S^2$. We can choose a set of coordinates $\{q^i\}$ to label the points in the configuration space, which are called generalized coordinates.

The physical states are not only the points in the configuration space, but also include the tangent vectors of the configuration space, which are called generalized velocities $\{\dot{q}^i\}$. Thus, the physical states of the system are points in the tangent bundle $T\mathcal{Q}$ of the configuration space $\mathcal{Q}$. For example, for a particle moving in three-dimensional space, the physical states are points in $\mathbb{R}^6$, which is the tangent bundle of $\mathbb{R}^3$. Here the tangent spaces at different points are all $\mathbb{R}^3$. We compare the generalized velocities at different points by directly subtracting them. For a rigid body, the tangent space is $\mathbb{R}^3 \times \mathfrak{so}(3)$, where $\mathfrak{so}(3)$ is the Lie algebra of $SO(3)$. To compare the generalized velocities at different points, we need to use the Levi-Civita connection.

[E1] The Lagrangian and the action functional

The Lagrangian L is a smooth function $L : T\mathcal{Q} \to \mathbb{R}$, which is denoted as $L(q^i, \dot{q}^i; t)$. It assigns a weight to each trajectory $\mathbf{q}(t)$ in the configuration space via the action functional:

$$S\{q^i(t)\} = \int_{t_i}^{t_f} L(q^i(t), \dot{q}^i(t); t)\mathrm{d}t. \tag{2.1}$$

[E2] The Hamilton principle and the Euler-Lagrangian equation

The physical trajectory of the system is the one that makes the action functional

stationary, i.e. the variation of the action functional vanishes:

$$\delta S = 0, \tag{2.2}$$

which is called the Hamilton principle. From the Hamilton principle, we can derive the Euler-Lagrangian equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0. \tag{2.3}$$

Now a natural question is: how to find the Lagrangian for a given system? There are two answers. The first one is to use our experience. The Lagrangian is the subtract of the kinetic energy and the potential energy, which is $L = T - V$. The second one is to use the symmetries of the system.

[D1] From symmetries to the Lagrangian of a free particle in three-dimensional space

We have the following symmetries for a free particle in three-dimensional space:

- Time translation symmetry: $t \to t + \Delta t$.
- Space translation symmetry: $\mathbf{x} \to \mathbf{x} + \Delta\mathbf{x}$.
- Rotation symmetry: $\mathbf{x} \to R\mathbf{x}$, where $R \in SO(3)$.
- Galilean boost symmetry: $\mathbf{x} \to \mathbf{x} + \mathbf{v}t$.

The dynamics of the system should be invariant under these symmetries. The time translation symmetry requires that the Lagrangian does not depend on time t explicitly. The space translation symmetry requires that the Lagrangian does not depend on the position $\mathbf{x}$. The rotation symmetry requires that the Lagrangian takes the form $L = L(\dot{\mathbf{x}}^2)$. The Galilean boost symmetry requires that

$$L((\dot{\mathbf{x}} + \mathbf{v})^2) - L(\dot{\mathbf{x}}^2) = L(\dot{\mathbf{x}}^2 + \mathbf{v}^2 + 2\mathbf{v}\cdot\dot{\mathbf{x}}) - L(\dot{\mathbf{x}}^2) \tag{2.4}$$

is a total time derivative. Thus, we have $L \propto \dot{\mathbf{x}}^2$.

Chapter 3. Hamilton mechanics

[E1] Possible states in Hamiltonian mechanics

The physical states in Hamiltonian mechanics are points in the cotangent bundle $T^*\mathcal{Q}$ of the configuration space $\mathcal{Q}$, which is called the phase space. The cotangent bundle is the dual space of the tangent bundle. We denote the coordinates of the cotangent bundle as $\{q^i, p_i\}$, where p_i are called the canonical momenta.

[E1] Symplectic structure for the phase space

There exists a natural symplectic structure on the cotangent bundle, which is given by the symplectic form

$$\omega = \mathrm{d}q^i \wedge \mathrm{d}p_i, \tag{3.1}$$

which is a closed non-degenerate 2-form.

[D1] From function to trajectory in symplectic space

We may ask a question: why do we need symplectic structure? Recall that in the Lagrangian mechanics, we get the physical trajectory from the Lagrangian function (and the boundary conditions, always), which is a map from the physical state space to the real numbers. Here we also want to get the physical trajectory from a function from the physical state space to the real numbers. The physical state space in Hamiltonian mechanics is the phase space. So the question becomes: in a manifold, how to get a trajectory from a function? We know that the function can define a differential 1-form by the exterior derivative,

$$\mathrm{d} : M \to T^*M, \quad f \mapsto \mathrm{d}f, \quad \mathrm{d}f(X) = X(f). \tag{3.2}$$

And the trajectory $\gamma_i(t)$ can be generated by a vector field (with boundary conditions) by

$$\frac{\mathrm{d}\gamma_i}{\mathrm{d}t} = X_i(\gamma(t)), \quad \gamma_i(0) = \gamma_{i0}. \tag{3.3}$$

Then the mystery becomes

$$f \to \mathrm{d}f \xrightarrow{?} X_f \to \gamma_f(t), \tag{3.4}$$

i.e. how to get a map from the 1-form to the vector field. Symplectic structure can do this job. First, the symplectic form ω can define a map from the tangent space to the cotangent space by

$$X \mapsto \iota_X \omega, \quad \iota_X \omega(Y) = \omega(X, Y), \quad X \in TM, \forall Y \in TM. \tag{3.5}$$

Since ω is non-degenerate, the inverse map exists, which means for a given 1-form α, there exists a unique vector field X_α such that

$$\iota_{X_\alpha} \omega = \alpha. \tag{3.6}$$

We can compute it in the local coordinates. We write $\alpha = \alpha_i \mathrm{d}q^i + \beta^i \mathrm{d}p_i$ and $X_\alpha = X^i_\alpha \frac{\partial}{\partial q^i} + Y_{\alpha i} \frac{\partial}{\partial p_i}$. Then we have

$$\iota_{X_\alpha} \omega = X^i_\alpha \mathrm{d}p_i - Y_{\alpha i} \mathrm{d}q^i = \alpha_i \mathrm{d}q^i + \beta^i \mathrm{d}p_i. \tag{3.7}$$

Thus, for the function $f(q^i, p_i; t)$, we have $\mathrm{d}f = \frac{\partial f}{\partial q^i} \mathrm{d}q^i + \frac{\partial f}{\partial p_i} \mathrm{d}p_i$. The corresponding vector field is

$$X_f = \frac{\partial f}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial f}{\partial q^i} \frac{\partial}{\partial p_i}, \tag{3.8}$$

which is called the Hamiltonian vector field. And the flow generated by the vector field X_f is given by

$$\frac{\mathrm{d}q^i}{\mathrm{d}t} = \frac{\partial f}{\partial p_i}, \quad \frac{\mathrm{d}p_i}{\mathrm{d}t} = -\frac{\partial f}{\partial q^i} \tag{3.9}$$

Especially, for the Hamiltonian function $H(q^i, p_i; t)$, the trajectory generated by the Hamiltonian function is the physical trajectory of the system, which is given by the Hamilton's equations of motion (or called canonical equations):

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}. \tag{3.10}$$

The Poisson bracket

$$\{f, g\}_{\mathbf{q},\mathbf{p}} \equiv \sum_{j=1}^{S} \left(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \right). \tag{3.11}$$

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, H\}_{\mathbf{q},\mathbf{p}} + \frac{\partial f}{\partial t} \tag{3.12}$$

$$\{p_i, p_j\}_{\mathbf{q},\mathbf{p}} = \{q_i, q_j\}_{\mathbf{q},\mathbf{p}} = 0, \quad \{p_i, q_j\}_{\mathbf{q},\mathbf{p}} = \delta_{ij} \tag{3.13}$$

Chapter 4. Hamilton-Jacobi theory